

Verification and readiness

MathGov/RippleLogic v13.0 · MG-RL-13.0-20260923-RELEASE-G · 23 September 2026

Acceptance scope

Accepted for the declared Tier 1–3 research-and-teaching specification, bounded frozen worked example, complete reading copies and supplied scoped reference tests. Not a general operational-selector, empirical welfare-validation, full canonical machine-registry, production assurance or domain execution certificate.

Independent baseline and patch evidence

The exact supplied 277-entry ZIP passed CRC and its prior public verification suite before editing. The copied current build contains the approved patches only, followed by necessary release-identity and reading synchronization. The exact prior ZIP is separately preserved.

Current prepackage suite: all 17 substantive commands passed (the final hash-ledger command runs only after freeze). Unit tests: 273. Independent workbook arithmetic: 900/900. Full formula replay: 20,212/20,212, with zero cached formula dependencies used. Workbook publication assertions: 55/55. External mutations: 12/12. Stored-formula probes: 10/10. Retained correction suites: 80/80, 77/77, 74/74, 290/290 and 857/857. New micro-patch assertions: 91/91. Static numbered navigation: 125/125. Reading/source/link checks: 791/791. Dashboard: 15/15. The final delivery receipt records the rerun of the completed final ZIP, including actual source/link-check totals, rather than assuming these prepackage counts remain identical after publication-page preparation.

Native application acceptance on the new workbook

Engine: LibreOffice 25.2.3.2 520(Build:2). Exact input SHA-256:
f6476e9192004fca887520d1144475bb62a7659ac7950b01d1c916882d7c2a5b.

Hard recalculation: 20,212 formula-result comparisons, zero mismatches and no tested formula errors. Acceptance/mutation checks: 23/23. A saved test copy was closed and reopened in a fresh Calc process; all 20,212 original formula results matched again with zero differences or errors. Absolute and relative tolerance: 1e-12. Core workbook unchanged by testing.

The Calc-reserialized XLSX is an evidence copy, not the master. Native serialization can change formula spelling or convert Boolean literals into equivalent formulas. This is disclosed and not used to rewrite the format-preserved source. The saved evidence copy is retained privately in the Complete archive. Reproduction utility: Native_Verification/verify_native_roundtrip.py. No Microsoft Excel parity was executed.

Numerical preservation

All 20,212 formula texts and their original cached results have the same fingerprint as the baseline. Only 20 literal cells changed (ten principal labels/records and ten exact snapshot mirrors). $RLS(A)=0.0153129427168233$; $RLS(B)=-0.00112238738505248$. $CVaR(A)=0.0219$; $CVaR(B)=0.0315$. The final framework verdict remains REFUSE_DETERMINISTIC_SELECTION, with authority choice separate.

Format and navigation

All 14 DOCX containers are readable; their supporting ZIP parts, inherited table properties and section geometry were compared. All 336 inherited table-format structures were preserved. All 95 XLSX sheets retain order and geometry. Complete PDF rendering totals 686 pages. Comparing current and baseline pages at 72 dpi found 533 identical pages and 153 changed pages due to authorized text flow and current identity. Changed-page contact sheets and the amended pages were reviewed; no new clipping or overlap was observed. Automated scan: zero blank pages and zero out-of-page text blocks. This is not exhaustive independent human proofreading.

The 70 Canon, 28 SGP and 27 WDBIP numbered opening references match their actual PDF hyperlink destinations and visible labels. Three Canon labels changed. No live PAGeref or auto-rewrite mechanism was introduced. Word may repaginate on a different host; the packaged PDF is the reference pagination.

Independent-review limits

The changes were derived from a supplied external LLM audit, adjudicated and independently recalculated within this build environment. This is not an independent human panel, field experiment, measurement-invariance study, full production runtime/security review or ProofPack. Claims depending on absent schemas/registries, WDBIP executable parity or RPAP/PFAP must satisfy their separate requirements. No live GitHub/site upload, hosted Actions run or interactive-browser acceptance was performed. Static links/body projections and local JavaScript tests were checked.

Evidence ownership and hash-last freeze

Verification/Final_G/ identifies current execution/native/render evidence. Earlier retained named release records are historical and do not certify changed bytes. Original tests remain; two older exact-whole-workbook assertions were adapted to current identity plus the invariant full formula/cache fingerprint, because approved text labels changed. They were not weakened to tolerate numerical drift.

After reports, projections, inventory and Core files are fixed, the component manifest and SHA-256 ledger are generated last. The delivery receipt outside the ZIP records final archives, hashes, fresh-extraction tests, native rerun and before/after inventory comparison. Any later file edit invalidates that exact frozen identity and needs a new build record.